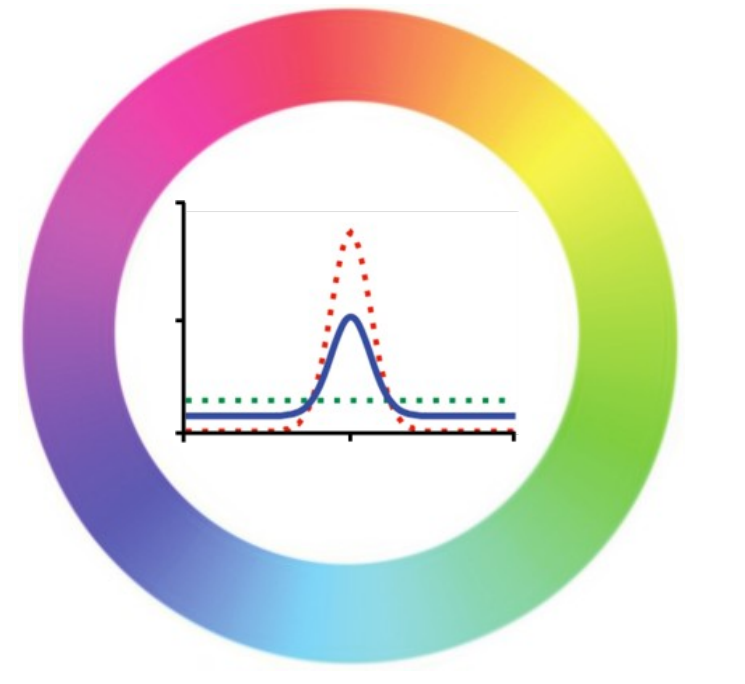




Revisiting Evidence Against the Discrete-Capacity Account of STM limits: Objective Guessing Obscured by Representational Modeling

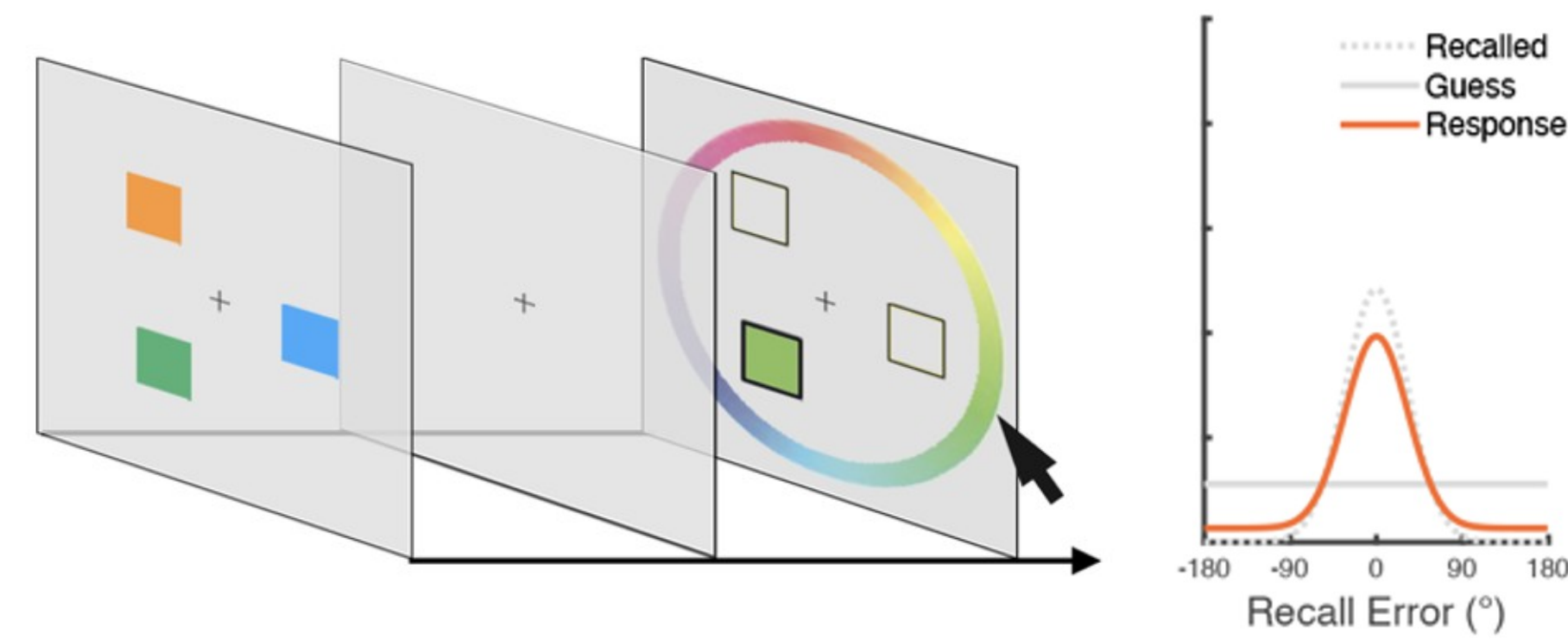


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Short-term memory (STM) is limited by ...

Continuous Resource: continuously divisible resource (e.g., a global pool of neural activity, or the total amount of information a system can process).

Discrete Capacity: a finite number of discrete items or “chunks” or “pointers”.



Prediction from the *Capacity* account:
Objective guessing during STM recall.

The *Mixture* model (Zhang & Luck, 2008):
Distribution of recall errors = Memory-driven responses + Guesses.

The *Target Confusability Competition (TCC)* model (Schurgin et al., 2020):
Low signal-to-noise ratio (d') => Fat tails in the error distribution. No guess.

TCC **outperformed** Mixture in model comparisons (e.g., based on BIC).

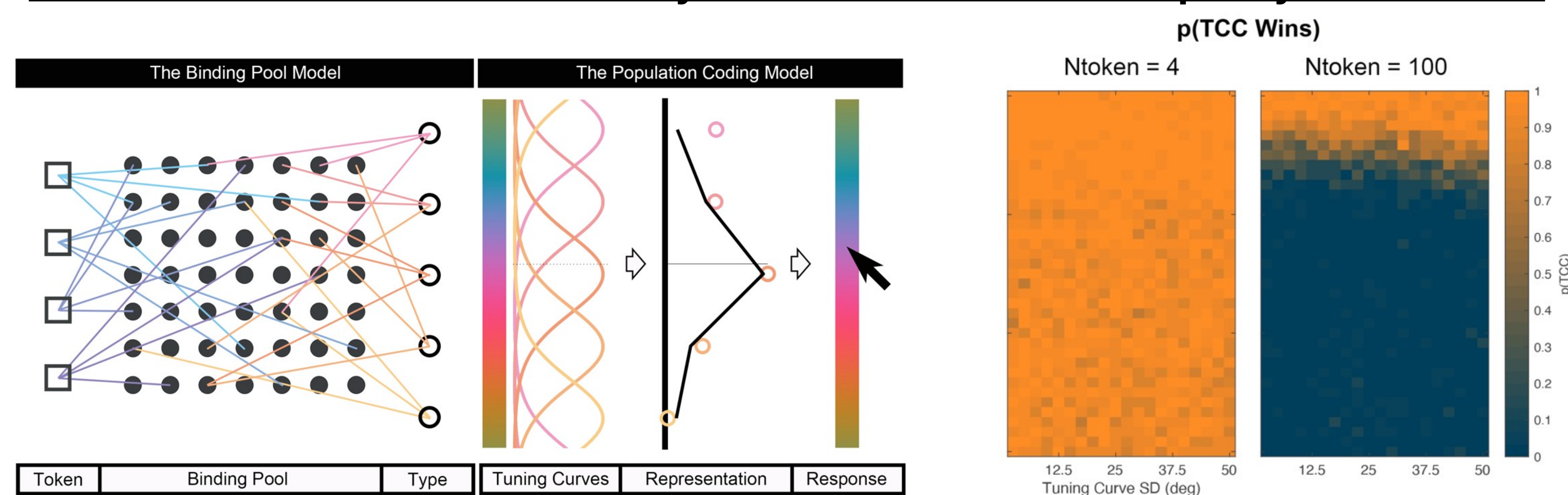
No objective guessing during recall?

No discrete capacity limit in STM?

The Mixture model is importantly wrong?

General question: Can the models identify objective guessing during STM recall?

Simulation: Mathematical Mimicry Obscures Simulated Capacity Bottlenecks

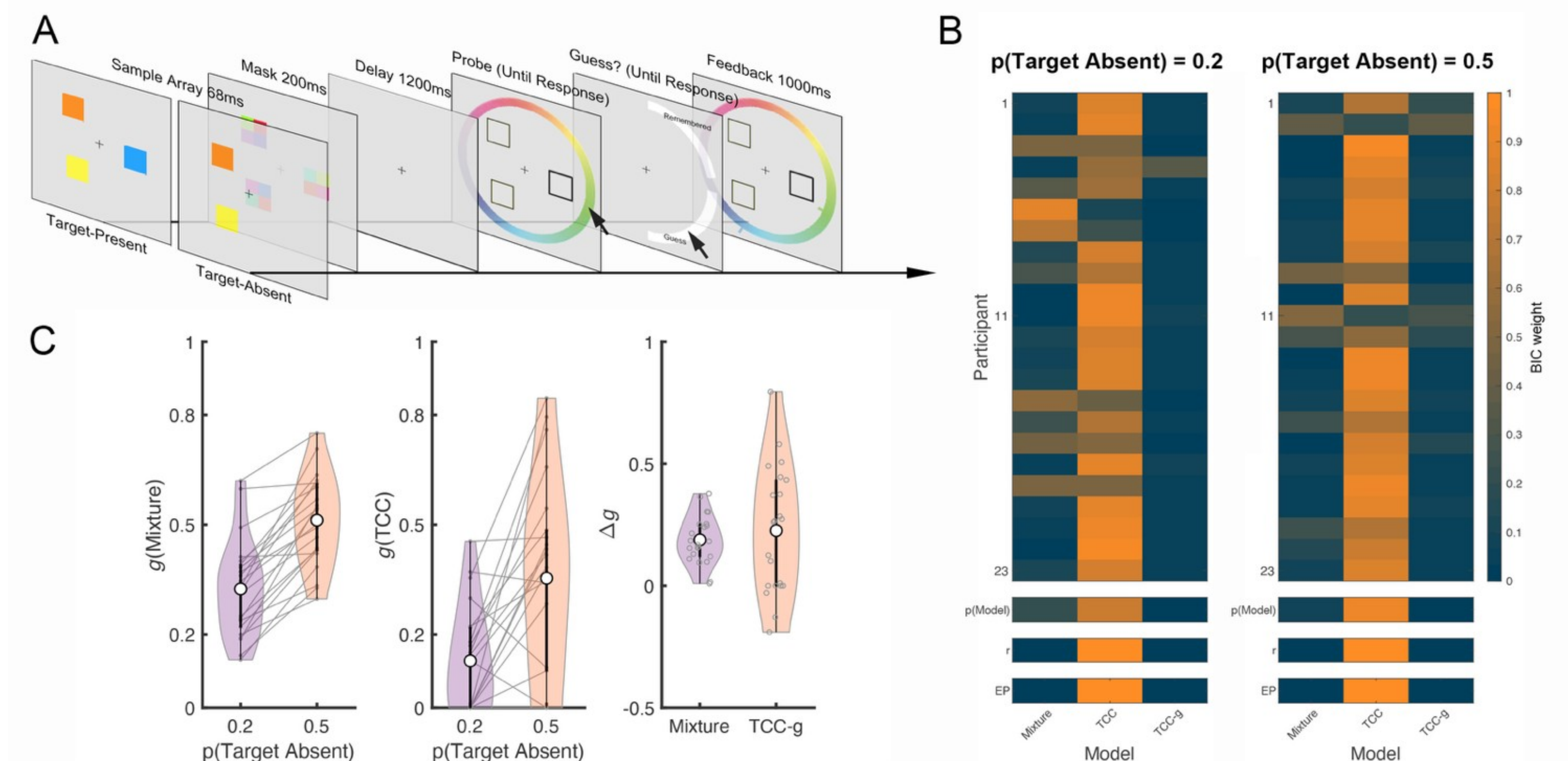


- Ground truth of STM limits simulated from a process model.
- TCC won the comparisons even when STM is primarily limited by a discrete capacity.

Empirical Study: "Forced Guessing" Reveals the Composite Nature of d'

Task: Continuous color recall with 20%/50% target absence probability.

- Ground truth of guessing established from the experimental manipulation.
- TCC still won the model comparisons in both conditions.
- Guess rate measured by the “TCC + Guess” model is lower than the “ground-truth” lower bound of guess rate.

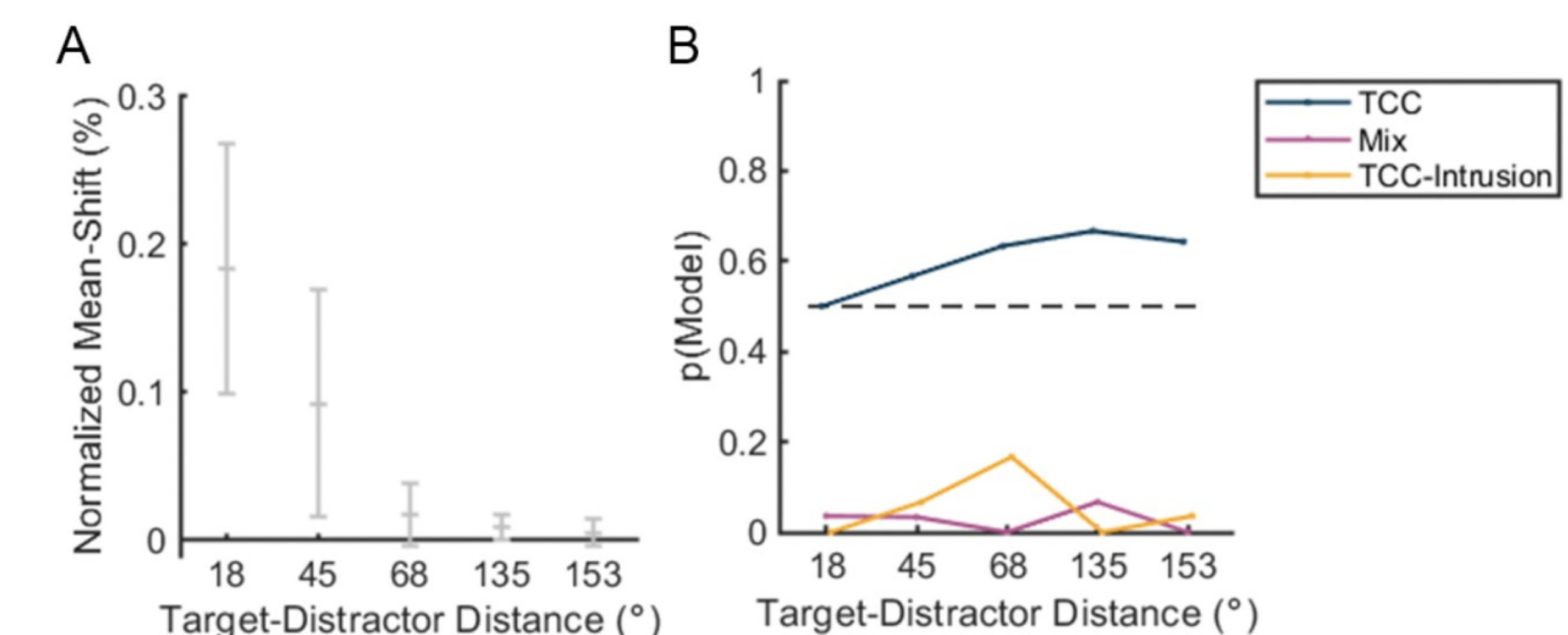


Re-Analysis: Continuous Metrics Over-Explain Variance Arising from the Intrusion of Perceptual Distractors During Memory Maintenance

Data: Zhang & Lewis-Peacock, 2025, *Psychological Science*

Task: Continuous recall of orientation with perceptual distraction during memory maintenance.

- TCC still outperformed the TCC-Intrusion model (and all other models with explicit components capturing the effect of perceptual distraction).



Discussion

- Model comparisons with BIC failed to detect the existence of objective guessing during recall.
- TCC d' can over-explain variances arising from processes not explicitly modeled by TCC (e.g., objective guessing, bias from perceptual distraction).
- Statistical supremacy $\neq$ Theoretical validity.
- How do we evaluate/compare theories through cognitive modeling?